

Topic 09: Exact Differential Equations & Integrating Multipliers

Professor Leonard Master Series · Lessons 28 to 30

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Topographic Mountain

"Imagine a 3D mountain whose height above sea level is $F(x, y)$. If you walk along a level hiking trail that stays at the exact same elevation (say, $F(x, y) = C$), your change in elevation is ZERO: $dF = 0$."

In multivariable calculus, the total differential of $F(x, y)$ is:

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = 0$$

Now look at a differential equation written in differential form:

$$M(x, y)dx + N(x, y)dy = 0$$

If this matches $dF = 0$, then:

$$M(x, y) = \frac{\partial F}{\partial x} \quad \text{and} \quad N(x, y) = \frac{\partial F}{\partial y}$$

By Clairaut's Theorem on equality of mixed partial derivatives:

$$\frac{\partial}{\partial y} \left(\frac{\partial F}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial y} \right) \iff \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

That is the **Exactness Test!** If $M_y = N_x$, there is a single potential function $F(x, y)$ whose level curves $F(x, y) = C$ are the solutions to the differential equation!

2. Leonard's 4-Step Whiteboard Protocol

1. **Test Exactness:** Compute $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$. If equal, celebrate!
2. **Partial Integration:** Integrate $M(x, y)$ with respect to x :

$$F(x, y) = \int M(x, y) dx + g(y)$$

(Note: The integration constant is an unknown function of y , $g(y)$!)

3. **Differentiate with respect to y :**

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y} \left[\int M dx \right] + g'(y)$$

Set this equal to $N(x, y)$ and solve for $g'(y)$.

4. **Integrate $g'(y)$ & State Final Implicit Solution:** Integrate $g'(y)$ to find $g(y)$. Write the answer as:

$$F(x, y) = C$$

3. What if it is NOT Exact? (Finding the Magic Multiplier)

If $M_y \neq N_x$, we seek a multiplying factor μ :

- **Case 1:** If $\frac{M_y - N_x}{N}$ depends **only on x** , then:

$$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}$$

- **Case 2:** If $\frac{N_x - M_y}{M}$ depends **only on y** , then:

$$\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}$$

Multiply the original equation through by μ . It is now guaranteed to be exact!